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Retaining students at Aalborg University

How are we successfully retaining students? Case: Interaction Design (IxD) 2014–2019.

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ABSTRACT

Dropouts from higher educations have been high for several years. As increasingly within the research field, the focus in this project is the positive angle of the problem – 'retention'. The study case is the Interaction Design (IxD) program, a study doing well regarding retention. The research question is: *What is it that the IxD program does well regarding retention?*

We have used an explanatory mixed methods approach, as we have used both qualitative and quantitative research methods. Qualitative data was used to design the questionnaire, and to substantiate the quantitative results. The questionnaire was send out to all students from first until 7th semester. Alongside this, we also interviewed staff.

In terms of state of the art within the research area of retention, this piece of work highlights the importance of expectation reconciliation amongst different stakeholders.

INTRODUCTION – RESEARCH QUESTION AND METHODOLOGY

Dropout rates from higher education have been high for several years. The dropout percentage has been 16% for the last 10 years for first-year students; this indicates a problem (1). Previously Aalborg University had a low dropout rate compared with other universities. However, during the last few years an increased dropout rate among first-year students at Aalborg University has given rise to concern. There are great differences in rates for dropouts when one compares different programmes. Knowledge about the causes of the problem is essential. Another perspective which can shed light on why there are differences between the different programmes and why some programmes are clearly better than others is just as essential. During the

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last few decades, focus on the problem has changed from a negative approach – 'dropout' – to a positive approach – 'retention' (2, 3). The focus of this study is on the positive angle of the problem: 'retention'. The case study used is the Interaction Design (IxD) programme, which has been performing well with regard to retention. The research question is:

What does the IxD programme do well regarding retention?

As a starting point for qualifying our research, questions, and establishing a focus in relation to our empirical collection and data analysis, the theory of a 'sense of belonging' was used (2, 3). This research shows that successful retention is linked to the student's experience of belonging ('sense of belonging') to the education, professionally as well as socially. In other words, the students' perception of being academically accepted, appreciated, and included, and their experience of being an important part of social life and social activities, is important to the retention of students. Other research (1, 4) points out that the start-up phase and the first year are particularly critical for retention. We have chosen, however, to involve students from the first semester up to and including the seventh semester, because we know from the outset that the older students have a great influence on the academic and social environment of the entire programme. Based on this and the above comments on the importance of the context, the following three sub-questions were identified:

What type of students attends the education programme and how do they experience the programme's start-up phase?

How does professional and social integration look at each four-year stage: i.e., the first, third, fifth, and seventh semesters?

In what institutional context does the IxD programme unfold, especially with regard to the infrastructure implemented by the supervisor (IT, group rooms, secretaries, etc.) and student organizations?

1.1 Method for data collection

We used an explanatory mixed methods approach (5), because we used both qualitative and quantitative research methods. In our quantitative approach, a questionnaire was sent to all students in the first to seventh semesters. The qualitative data were used to design the questionnaire and to support the quantitative results. The qualitative data consist of three focus group interviews with, respectively, five students from the first semester, seven students from semesters three to seven, and two teachers and supervisors who helped to found the programme. In addition, the student counsellor for the programme was interviewed. Data on dropout rates were collected for the individual semesters. Data from the questionnaire and interviews were subsequently analysed according to the three sub-questions. In the following paragraphs, the results of the analysis are first reviewed in a number of sections, after which the main question of the study is discussed.

The questionnaire was sent to 129 students from semesters one through seven. Responses were received from 55 students: 22 women and 33 men. Thirty-six students answered all the questions. Replies were received from 19 students in the first semester, 13 students in the third semester, 9 students from the fifth semester,

and 11 students from the seventh semester (see Table 1). We consider the answer rate satisfactory and have no reason to assume that the responses are biased.

Table 1. Response rate for questionnaire survey

Semester	Number of questionnaires	Number of answers	Answer Rate
1st semester	37	19	51.4%
3rd semester	37	13	35.1%
5th semester	39	9	23.1%
7th semester	16	11	68.8%
Total	129	55	42.6%

2 BASIC DATA, EXPECTATIONS, AND START-UP

2.1 Satisfaction with the programme

The study results show that 33% of the students are very satisfied with the programme and 53% are satisfied with the programme. No student is very dissatisfied with the programme.

On the question of satisfaction, there is a difference in the answers between the first to third semesters and fifth to seventh semesters. The first to third semesters answered that 47% are very satisfied and 47% are satisfied. For the students in the fifth to seventh semesters, 7% are very satisfied and 67% are satisfied. There are relatively more very satisfied students in the earlier semesters.

2.2 Motivation for choosing study and start-up

A number of questions about what influenced the students' choice of programme reveals that design and creativity, problem-solving, construction of new solutions, and technology have great influence or some influence on their decisions, whereas the expectations of others and tradition have no influence or limited influence.

If we look specifically at how the students experienced starting at Aalborg University, there is great satisfaction with their experience of the introduction period. The students were asked how they were received at Aalborg University, how the tutor scheme worked, and how the introductory period progressed. Everyone replied that these were very good or that they were well received. For first to third semester students, 43% were very satisfied, while for fifth to seventh semester students, 33% were very satisfied. Regarding the question of the tutor scheme, 92% of first to third semester students responded that it works very well or well. Eighty-four percent of fifth to seventh semester students answered that it works very well or well. With regard to the introductory course, 93% of first to third semester students answered that it works very well or well. Eighty-four percent of fifth to seventh semester students answered that it works very well or well. A comparison of the answers for the different semesters – first to third semesters and fifth to seventh semesters – shows that the first to third semester students are generally the most positive. The students generally agree that the tutors make a fantastic effort. It was also mentioned that the professional start-up

is quiet and calm. The students also commented on what topics could be improved in the on-boarding. In particular, they mentioned that more information before the start is desired.

3 PROFESSIONAL INTEGRATION

Professional integration is very important for the students' ability to retain the knowledge of the subject they have chosen. We asked the students what aspects of study are important to them. They responded broadly that the subject's interest, relevance, usability, quality, and learning environment and balance between study and leisure are the most important parameters, together with a good social environment and the capability to finish their studies. A few quotes from first to third semester students illustrate some of these points:

> *…that I can see a relevance in what I am taught and can use it now in the semester project, but also in the future. In addition, it is also important to have a good study environment, both among one's fellow students on the programme, but also from other study programmes.*

> *An educational environment where there is also the opportunity to be with my fellow students socially, a good education that interests me, good teachers, good literature, a group room for the project group.*

Similar comments were recorded from the students in semesters five to seven; however, they added parameters such as a correspondence between expectations and realities, student involvement, a connection between projects and courses, and good facilities. They also pay attention to the importance of the size of the classes. In relation to professional integration, it was emphasized that it is important that the classes do not become too small (i.e., under 30 students) for the sake of the possibility of entering into different types of academic relations with fellow students and in terms of providing a better opportunity to adapt courses specifically to their professional skills. On the other hand, the classes should not be too big because, in this scenario, social integration is compromised. Students from the later semesters also emphasized the importance of the projects being realistic and the existence of a professional challenge.

3.1 Workload, project work, and coherence of courses

With regard to the workload, 75% of both the younger and older students experienced it as comfortable, while for 20% it was too much. Seventy percent of the students in semesters one to three found the degree of difficulty comfortable, while 26% found it difficult or very difficult. In the fifth to seventh semesters, 50% of students found the degree of difficulty comfortable and 33% found it difficult. The students experienced an increase in the level of difficulty in the later semesters.

We asked the students to list good aspects about studying at Aalborg University. The common answer across all semesters was group work and Aalborg University's basic PBL model. A few examples are as follows:

> *I really like the group work that I know helps me to prepare for my future in the labour market.*

> *Group work, PBL, Aalborg as a study city, the people.*

This fits well with the feedback on how important it is for the individual student to work in project groups, which was 89% and 92% for first to third semesters and fifth to seventh semesters, respectively. This is also reflected in how important it is for the individual student that projects are perceived to be relevant. For 100% of the students in the fifth to seventh semesters, this is of great importance, while it is of great importance to 78% of students in the first to third semesters.

Respectively, 74% and 67% of first to third semester students and fifth to seventh semester students found that the relevance of the project work they are participating in is great. Thus, there is better correspondence between expectation and reality in the first to third semesters than there is in the fifth to seventh semesters. The focus group interviews with the students also show the importance of the projects. The following quotation is from students in the fifth to seventh semesters:

Projects work well!

A relationship that traditionally has an impact on academic integration at Aalborg University is the coherence between courses and projects. Ninety-seven percent of students in the first to third semesters think there is coherence or a very large coherence. The tendency is for a downward trend in the fifth to seventh semesters, where 67% think there is coherence or a very large coherence and 33% see only a little coherence.

3.2 The professional environment and the students' own initiatives

The big picture shows that 88% of the students surveyed think that there is a good to very good academic environment for their education, 80% believe that they are part of the academic environment, and over half of the students in both groups believe they have an influence on their professional environment. At IxD, a student-driven professional club called FixD has been established, which has been a driving force in establishing professional and social activities. This group has had a significant influence on the academic environment. FixD and its activities are more pronounced in the fifth to seventh semesters.

4 SOCIAL INTEGRATION

Social integration, together with professional integration, is important for retaining students. The issues related to social integration are the social environment, a sense of belonging, and the students' own initiatives.

4.1 Social environment

Regarding the overall question of how the social environment works, 92% of first to third semester students answered that it works very well or well. For fifth to seventh semester students, 75% responded that it works very well or well.

The student initiative FixD is of great importance to the social field, and it has a little more importance for the older students than for students in the initial semesters. For the first to third semesters, the student initiative is of great importance to the social environment for 46% and has little importance for 31%. Twenty-three percent indicated that they did not know. For fifth to seventh semester students, the student initiative

has a great importance for 67%, a little significance for 17%, and no significance for 17%.

The experience of being part of the social environment is greatest among fifth to seventh semester students. Here, 66% experience being part of the social environment to a great extent and to some extent. For first to third semester students, just 47% experience being part of the social environment to a great extent or to some extent. Forty-six percent of first to third semester students feel to a lesser extent part of the social environment; the corresponding group of fifth to seventh semester students account for 25%.

5 PROFESSIONAL AND SOCIAL INTEGRATION – SENSE OF BELONGING

5.1 On a scale of one to ten, how much do you feel that you belong here?

The students have been asked to indicate, on a scale from one to ten, how much they feel that they belong to the programme. There is a big difference in the experiences of belonging between the different semesters. First semester students experience a very high degree of belonging, compared with other semesters. The lowest affiliation is that of third semester students.
We asked about students' identity in both interviews and, already in the first semester, they have a clear and simple feeling of who they are and what their role will be in a professional context:

> *I have to learn to make a computer easier to use.* (First semester student)

> *We are natural born problem solvers.* (Fifth to seventh semester students)

6 INSTITUTIONAL FACTORS

The institutional factors deal with the supervisory body, collaboration with the study secretary, IT structure, IT support, scheduling, teaching platform, and premises, etc. We began by looking at the programme's establishment and development before moving on to other institutional factors.

6.1 The establishment and development of the programme

IxD is a new programme that started in 2014. The first candidates will complete the programme in the summer of 2019. The programme is small with an admission in 2018 of 38 students. According to interviews with two of the leading forces in the establishment of the programme, the background for its creation was both a need in the industry and a recognition of the importance of further developing the students' academic expertise in the field:

> *There are a lot of jobs that are about working with users…. There is a demand for someone who can build something… who can be constructive and not just creative. This is a design-oriented education,… a technical education.... That's what the industry demands.*

> *We designed the education programme in the way that we thought was the right way to approach this type of education.*

There is a clear and meaningful outside demand. The composition and content, and

the relationship between projects and courses, are also elements the supervisor group worked on diligently and intensely. The supervisors have been very careful to integrate well individual elements of each semester and then focus on explaining the programme to the students.

During the first few years, there were some adjustments. In order to be able to develop the education, it was important for the teachers to have 'a finger on the pulse' with regard to the students; in particular, the first year students' progress was followed closely. The teachers have been very responsive towards the wishes of the students and they have had a crucial role in initiating social activities:

> *Some of them [the social activities] we pushed and started, and then the students took over. And that's actually a dream situation.*

From the focus group interviews with the older students, it is clear that there is great initiative and ownership among the students in the later semesters. They are very committed to their education and very dedicated to each other. When there is something that does not work according to the students' wishes, they have a great drive to change things. This drive is coupled with a great deal of scepticism towards the university, the teachers, and the structures they are part of.

6.2 Nature and composition of the supervisor group

There is an internationally recognized research group at IxD. As mentioned, they have been very much involved in the establishment of the programme. There is great enthusiasm among the teachers and the education is their 'lifeblood', which means that they often 'run 10% longer':

> *It is very much driven by the fact that it has been very important for us to get this education both accredited and up and running.*

It also means a lot to the teachers' international reputations to have a programme in their own field of research that is very successful. Many resources from the teachers' side have been used to progress the programme to the level it is at now. Thus, two of the challenges of the programme are its success and its scalability. It is, for example, difficult to acquire newly qualified staff, so there are vacancies that are not occupied.

6.3 Education structure: scheduling

Eighty-two percent of the students in the first to third semesters answered that the education structure and scheduling work very well or well. For fifth to seventh semester students, the corresponding figure is 58%. One of the challenges that has emerged from the focus group interview with the older students is the merged classes they have with other programmes. The students feel overlooked when, for example, they study with Architecture & Design students.

> *It has happened several times that lecturers have forgotten that we are present. …[They have] not prepared anything for us.*

The teachers are aware of the problem and say:

> *I think that the merging… has had some challenges. We have changed a lot of this, but still have not quite found the right solution.*

The challenge is that there is a difference in the structure of the programmes. The idea of mixing programmes in courses is liked by the supervisors but, in practice, it can be challenging. It may look 'good on a slide', but the reality is more complicated.

6.4 Teaching platform and group rooms

Twenty-two percent of the students in the first to third semesters answered that the teaching platform works well. For fifth to seventh semester students, the corresponding figures are 58%. Of the students in the first to third semesters, 48% answered that the teaching platform works less well. For fifth to seventh semester students, 25% responded that the teaching platform works less well. Thirty percent of first and third semester students and 17% of fifth to seventh semester students responded that the teaching platform works poorly.

Own group rooms are of great importance to the students. Eighty-seven percent of all students state that it is very important to have their own group room. The same percentage of all students thinks that the group rooms work well or very well. When asked for a reason why it is important to have one's own group room, one of the student answers was:

> *You can do more concentrated group work and work without disturbances from others or alarms from other students. You can also have your own things there, especially if you are doing some sketches or other things that can be difficult to take back and forth; it is nice that it can be locked and no one can come and take it.*

7 DISCUSSION

In accordance with the initial identification of the design of the study, we will first answer the three sub-questions and then answer the main question of the study: what does the IxD programme do well in relation to retaining students? In conclusion, we will present the results of the analysis in order to transfer the experiences to other programmes at Aalborg University.

7.1 What type of students attend the programme and how did they experience the start-up phase?

When looking at the results of the analysis, it is clear that there is generally a great satisfaction with the IxD programme. This applies to all students. Even at the beginning, the students feel well received. They especially emphasize the tutors, who do a great job in connection at the start-up. An area that could be improved in connection with the start-up is the information that students receive before the start of their studies. Approximately half of the students state that they did not get enough information. In connection with the students' motivation for choosing the programme, it is clear that parameters such as design and creativity, problem-solving, construction of new solutions, and technology are important for the choice of the IxD programme. When interviewed, it became clear that the students have a very clear idea of what their study entails in terms of subject and identity, and they have a very clear understanding of their role as interaction designers. They identify themselves as interactions designers, even though it is a new education programme and they have a clear understanding that there is a need for precisely their qualifications in industry. The reason for this understanding and sense of identity may be due to a great deal of clarity from supervisors and teachers in terms of what is taught and what it can be used for.

7.2 How does professional and social integration look at the programme's first, third, fifth, and seventh semesters?

In relation to professional integration, the students at IxD mainly experience having an appropriate workload and an appropriate level of difficulty in courses and projects. Something that is widely agreed among the students is the importance of project work. Project work is the cornerstone of their education. It is one of the best things about going to Aalborg University. When project work is combined with great relevance and appropriate academic challenges, this is of great importance for professional integration. Another factor that has a great influence on professional integration is the coherence between courses and projects. There is a good coherence between the courses and the projects on the IxD program. The students also mention the students' own initiatives at FixD as an important part of professional integration. The efforts of the older students to help the younger students academically have great importance for professional integration. This, combined with a large satisfaction with supervisors and teachers, gives the students of the IxD programme good professional integration.

The social environment works well in the IxD program. Something that is very important to this is the students' FixD initiative, which organizes several social activities across the semesters. Many of the students consider themselves to be an important part of the social environment. All of these factors contribute to good social integration. The affiliation of the students with the IxD programme is considered to be relatively high. This is not least based on good professional and good social integration, but also based on the students' own assessments. The character of the students' affiliation is worth paying attention to. It seems as if the students in later semesters feel affiliated to the other students in the programme and hardly any affiliation with Aalborg University or the teachers. There seems to be a certain contradiction between the perspective of older students and Aalborg University as an institution, which is related to a large discrepancy between the students' visions and dreams of a very special education, where not only professionalism and the social environment are above average, but also the framework of the education. Overall the satisfaction from the 1st-3rd semester students are higher compared to the 5th-7th semester, which is evidence of a growing satisfaction with the education.

This development has been enhanced by the teachers being particularly responsive to what might not be working, as well as perhaps a lack of dissemination/recognition among the students about what claims a well-established university can generally meet and function. However, it is still the professional and social quality that is crucial for student retention.

7.3 In what institutional context does the IxD programme unfold?

In the institutional context in which the IxD programme unfolds, group rooms are essential. Other institutional factors, such as IT structure and IT support, collaboration with the study secretary, and lecture rooms work well. A challenge exists with merged courses, especially in the later semesters. Here, there are many cross courses with other programmes and this presents the challenge that IxD students often feel overlooked. This relationship is, among other things, because there is a structural difference between how the education is organized. Various solutions have been tried, but no optimal solution has yet been found. At the IxD programme, there is an internationally recognized research group. The mainstay of the IxD programme has been involved from the beginning. There is a great spirit among the teachers, which

rubs off on the students. This, however, is also the seed of a future challenge: to scale the education to the educational resources that are available for the programme. The supporting forces have been very involved in the development of the programme and with the growth of the programme it is necessary to recruit new teachers. This has so far been difficult; thus, there are several vacant positions among teachers and supervisors.

7.4 What does the IxD programme do well in terms of retaining students?

A summary of what the IxD programme does well is presented in Table 2 in a number of bullet points.

Table 2. What does the IxD programme do well, regarding retention?

The start-up	Motivation for choosing education
-Good reception -Good tutors -Good intro week	-Clarity on content and usability -Clear identity in relation to choice of programme
Professional integration	**Social integration**
-Appropriate workload and difficulty -Relevant projects -Good coherence between courses and projects -Good academic environment -Good guidance and good teachers -Student organization	-Good cohesion in the programme across the semesters -Student organization -The social environment
Institutional context	**Areas that could be improved**
-Own groups -Opportunity for collaboration across design spaces	-Information before start of study -Cross courses between programmes -The students' knowledge of and recognition of AAU's internal development logic

7.5 Perspectives on previous studies and transfer of experience

Previous studies at Aalborg University (6, 7) made it clear that, in other programmes, the expectation reconciliation with the students has not been sufficient, both in terms of rationale for the education in general and the rationality of the individual educational programme contents. The IxD programme directors are aware of the expectation reconciliation between students and university. When they organized the programme, it was based on a clear job-market-related rationale with a focus on high quality and well-founded progression. In addition, they were very attentive to communicating the idea and structure of the overall education and individual semester. This gives meaning and transparency to the students. In relation to what can be transferred to other programmes, we want to point out the following:

1. Expectation reconciliation with students is central. This is the key to a good collaboration between students and university.

2. The rationale behind the programme is important. Ensuring our programmes are in line with the demand of our students and the labour market is crucial for creating immediate logic and meaning.

3. The internal structure of the programme is of great importance. A focal point is to ensure an internal logic and trade-off between the different elements of a programme and a clear rationale behind which activities are placed when. In addition, it is important to at the same time take into account that the total workload does not become too large.

4. Agency and enthusiasts are crucial. The leading teachers who established the course have delivered an effort out of the ordinary and the motivation has come from within the programme. The same can be said about the students, even though the push towards collaboration has been largely due to the teaching staff's influence. Enthusiasts must take care not to be taken advantage of in order that they are able to both support their initiatives and ensure that they do not work over their abilities and run out of energy.

Of course, it is easier to apply points two and three in a new education programme. Similarly, it is probably easier to support the enthusiasts of a well-established programme, because it gives a lesser load to the same people.

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